

Conjecture 00000001144 is false:  
a hyperbolic Weyl group with growth rate below 2

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## 1 The conjecture and the reading

**Statement (English).** “Conjecture: The growth rate of hyperbolic-type Weyl groups of Kac–Moody algebras is 2 (hyperbolic Weyl growth).” The Chinese text says the same: the growth rate of the Weyl groups of Kac–Moody algebras of hyperbolic type is 2.

**Reading refuted.** For every hyperbolic generalized Cartan matrix (GCM)  $A$ , the Weyl group  $W(A)$ , with its standard generating set  $S = \{r_1, \dots, r_n\}$  of fundamental reflections, has exponential growth rate 2. We refute three versions of “growth rate”: the  $\limsup$  of  $a_k^{1/k}$ , where  $a_k$  is the number of elements of word length exactly  $k$ ; the  $\limsup$  of  $|B_k|^{1/k}$ , where  $B_k$  is the ball of radius  $k$ ; and the limit of either sequence. One explicit hyperbolic GCM, the matrix of the Feingold–Frenkel algebra  $\mathcal{F} = A_1^{++}$ , has both  $\limsup$ s at most 19/10; hence any limit of either sequence, if it exists, is at most 19/10, and in particular neither sequence tends to 2. (We do not prove that the limits exist; non-convergence to 2 already refutes the claimed value.)

**Readings not refuted.** (i) The existential reading, that *some* hyperbolic Weyl group has growth rate 2, is true. The rank-3 GCM with all off-diagonal entries  $-2$  is hyperbolic; numerical evidence (BFS up to a finite length, not a proof) suggests that its Weyl group has  $3 \cdot 2^{k-1}$  elements of length  $k$  for  $k \geq 1$ . (ii) For the readings “the growth rate is at most 2” and “the supremum is 2” we have only numerical evidence against them: for the simply laced rank-4 GCM whose diagram is the complete graph  $K_4$  (hyperbolic), finite BFS gives sphere ratios near 2.303. A finite BFS proves neither asymptotic convergence nor an all-length formula; these remarks are evidence only and are not used in, or proved by, the Lean disproof.

## 2 Definitions (as formalized)

**Definition 1** (GCM, indecomposable). A GCM is an integer matrix  $A = (a_{ij})$  with  $a_{ii} = 2$ ,  $a_{ij} \leq 0$  for  $i \neq j$ , and  $a_{ij} = 0 \iff a_{ji} = 0$  (Lean: `GCM`). The matrix  $A$  is indecomposable if no nonempty proper subset  $J$  of the indices has  $a_{ij} = 0$  for all  $i \in J$  and  $j \notin J$  (Lean: `Indecomposable`). Both definitions follow Wikipedia [W1].

**Definition 2** (Hyperbolic type, symmetrizable case). We use [W2]. Let  $A = DS$  with  $D$  diagonal with positive integer entries and  $S$  symmetric. Then  $A$  is hyperbolic if it is indecomposable,  $S$  is indefinite, and for each proper subset  $J$  of the indices the principal submatrix  $S_J$  is positive definite or positive semidefinite. Lean (`IsHyperbolic`) asks for: indecomposability; some  $d$  with  $d_i > 0$  and a real symmetric  $S$  with  $a_{ij} = d_i S_{ij}$ ; vectors  $x, y$  with  $x^T S x < 0 < y^T S y$ ; and  $x^T S x \geq 0$  for every  $x$  supported on a proper subset  $J$ . The last condition says exactly that  $S_J$  is positive semidefinite.

The text of [C] gives Kac’s general definition:  $A$  is hyperbolic if it is neither of finite nor of affine type, but every proper connected subdiagram is of finite or affine type. [W1] gives the minor

criteria: finite type if all principal minors are positive, affine type if the proper principal minors are positive and  $\det A = 0$ . For our matrix the Lean lemma `FF_minors` records the data for this definition as well. It gives  $\det A = -2$ , so  $A$  is neither finite nor affine. The connected proper subdiagrams are the three singletons  $(A_1)$ ,  $\{0, 1\}$  (minors 2, 3: finite,  $A_2$ ) and  $\{1, 2\}$  (minors 2 and  $\det = 0$ : affine,  $A_1^{(1)}$ ). The subset  $\{0, 2\}$  is disconnected because  $a_{02} = 0$ .

**Definition 3** (Weyl group, word length, growth rate). The reflection  $r_i$  acts on the root lattice  $\bigoplus_j \mathbb{Z}\alpha_j$  by  $r_i(\alpha_j) = \alpha_j - a_{ij}\alpha_i$ . In Lean, `reflMat` is the integer matrix of  $r_i$  in the basis of simple roots, and `refl` is the same map as a unit of  $M_n(\mathbb{Z})$ ; we prove  $r_i^2 = 1$ . The Weyl group `weyl` is the subgroup of  $GL_n(\mathbb{Z})$  generated by the  $r_i$ . Following [FKN], the Weyl group associated with a hyperbolic Cartan matrix is the group generated by the simple reflections. FKN write the reflections as  $w_I(X) = X - \frac{2(X, \alpha_I)}{(\alpha_I, \alpha_I)}\alpha_I$ . Our matrix is symmetric, so transposition conventions do not matter, and  $\det A \neq 0$ , so the simple roots are linearly independent. The ball `ball C k` is the set of products of at most  $k$  generators; since  $r_i = r_i^{-1}$ , words without inverses suffice. The sphere `sphere C k` is the set of elements in `ball C k` and in no smaller ball, i.e. of word length exactly  $k$ . Theorem `weyl_eq_iUnion_ball` proves  $W(A) = \bigcup_k B_k$ . The growth rate is `growthRate` =  $\limsup_k |\text{sphere}_k|^{1/k}$ . By Cauchy–Hadamard this is the inverse of the radius of convergence of the growth series  $\sum_k a_k t^k$ , which is how [KP] defines it. We also define `ballGrowthRate` =  $\limsup_k |B_k|^{1/k}$ , with  $B_k$  the ball of [W3].

Two conventions: `Set.ncard` is applied only to sets we prove finite, and the  $\limsup$  in  $\mathbb{R}$  is applied only to sequences we prove eventually bounded above and bounded below by 0, so neither takes a junk value.

### 3 The counterexample

The Feingold–Frenkel algebra is  $\mathcal{F} = A_1^{++}$ . According to [FKN], the discovery that its Weyl group is  $PGL_2(\mathbb{Z})$  was the first vital step in Feingold and Frenkel’s work. In the order  $\alpha_{-1}, \alpha_0, \alpha_1$  (Lean indices 0, 1, 2) its GCM is

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix} \quad (\text{ffA}).$$

This is the over-extension of  $A_1^{(1)}$ : the new node  $\alpha_{-1}$  is joined to the affine node  $\alpha_0$  by a single edge. The Lean proof does not depend on the identification: hyperbolicity is proved directly.

**Lemma 4** (`FF_hyperbolic`). *A is hyperbolic. Take  $D = I$  and  $S = A$ . The form is  $Q(x) = 2x_0^2 + 2x_1^2 + 2x_2^2 - 2x_0x_1 - 4x_1x_2$ . It satisfies  $Q(1, 2, 2) = -2 < 0$  and  $Q(1, 0, 0) = 2 > 0$ . If  $x_0 = 0$  then  $Q = 2(x_1 - x_2)^2$ ; if  $x_1 = 0$  then  $Q = 2x_0^2 + 2x_2^2$ ; if  $x_2 = 0$  then  $Q = x_0^2 + x_1^2 + (x_0 - x_1)^2$ . Indecomposability is checked by `decide`.*

**Lemma 5** (`FF_comm, refl_mul_self`).  $r_i^2 = 1$  for all  $i$ , and  $r_0r_2 = r_2r_0$ .

Call a word reduced if it contains neither  $ii$  nor the factor 20 (Lean: chains for the relation `okFF`).

**Lemma 6** (`ff_prepend, ff_reduce`). *Every word  $l$  has a reduced word  $w$  with  $|w| \leq |l|$  and the same product in  $W(A)$ .*

*Proof.* Induct on  $l$  and prepend one letter  $x$  to a reduced word  $w = yv$ . If  $xy$  is allowed, take  $xyv$ . If  $x = y$ , cancel. Otherwise  $x = 2$  and  $y = 0$ , and  $20v = 02v$ . If  $v$  starts with 2, cancel to  $0v'$ ; this is reduced, because the letter after 2 in  $v$  is neither 0 nor 2. If  $v$  starts with 1, take  $021v'$ . If  $v$  is empty, take  $02$ .  $\square$

**Lemma 7** (`ff_follow_card`, `ff_ball_card`). *The number of reduced words of length  $m+1$  that follow a letter  $a$  is at most  $v_a(5/3)^{m+1}$ , with  $v = (5, 5, 3)$ . The induction uses  $3(v_1 + v_2) \leq 5v_0$ ,  $3(v_0 + v_2) \leq 5v_1$  and  $3v_1 \leq 5v_2$ . Hence at most  $13(5/3)^j$  reduced words have length  $j$ , and  $|B_k| \leq 13(k+1)(5/3)^k$ ; in particular  $B_k$  is finite.*

**Theorem 8** (`FF_rate`, `disproof`). *For all large  $k$ , both  $|\text{sphere}_k|^{1/k}$  and  $|B_k|^{1/k}$  are at most  $19/10$ . Indeed  $13(k+1) \leq (57/50)^k$  eventually, by `tendsto_pow_const_div_const_pow_of_one_lt`, and  $(57/50)(5/3) = 19/10$ . Hence `growthRate FF`  $\leq 19/10$  and `ballGrowthRate FF`  $\leq 19/10$ ; neither sequence of  $k$ -th roots tends to 2; and*

$$\neg (\forall n \forall C : \text{GCM } n, \text{IsHyperbolic } C \rightarrow \text{growthRate } C = 2).$$

Remarks, not used in Lean. Breadth-first search on the reflection matrices up to a finite length gives sphere ratios close to  $1.3247\dots$ , the real root of  $x^3 - x - 1$  (numerical evidence only; a finite search does not prove convergence). This is consistent with  $W \cong PGL_2(\mathbb{Z}) \cong T(2, 3, \infty)$ , since  $(r_0 r_1)^3 = 1$  also holds. The rank-2 hyperbolic GCM  $\begin{pmatrix} 2 & -3 \\ -3 & 2 \end{pmatrix}$ , of which [W2] says there are infinitely many, has the infinite dihedral group as Weyl group: its spheres have size 2, so its growth rate is 1.

## 4 Lean formalization and axiom audit

File `lean/Conjecture1144/Basic.lean` (414 lines, `import Mathlib` only, Lean v4.33.1 with Mathlib v4.33.1). Main theorem:

```
theorem disproof :
  IsHyperbolic FF /\ FF.A.det = -2 /\
  (weyl FF : Set _) = (Union k, ball FF k) /\
  RateBound FF /\ growthRate FF != 2 /\ ballGrowthRate FF != 2 /\
  ~ (forall (n : Nat) (C : GCM n), IsHyperbolic C -> growthRate C = 2)
```

(ASCII transliteration of the Lean statement.) `RateBound FF` consists of: every ball is finite; both  $k$ -th-root sequences are eventually at most  $19/10$ ; both limsups are at most  $19/10$ ; and neither sequence tends to 2. The command `#print axioms C1144.disproof` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`; `decide` is used only on finite statements about explicit  $3 \times 3$  integer data.

## References

- [W1] Wikipedia, “Cartan matrix”, section on Lie algebras and classification. [https://en.wikipedia.org/wiki/Cartan\\_matrix](https://en.wikipedia.org/wiki/Cartan_matrix)
- [W2] Wikipedia, “Kac–Moody algebra”. [https://en.wikipedia.org/wiki/Kac%E2%80%93Moody\\_algebra](https://en.wikipedia.org/wiki/Kac%E2%80%93Moody_algebra)

- [W3] Wikipedia, “Growth rate (group theory)”. [https://en.wikipedia.org/wiki/Growth\\_rate\\_\(group\\_theory\)](https://en.wikipedia.org/wiki/Growth_rate_(group_theory))
- [C] L. Carbone, S. Chung, L. Cobbs, R. McRae, D. Nandi, Y. Naqvi, D. Penta, *Classification of hyperbolic Dynkin diagrams, root lengths and Weyl group orbits*, arXiv:1003.0564. <https://arxiv.org/abs/1003.0564>
- [FKN] A. J. Feingold, A. Kleinschmidt, H. Nicolai, *Hyperbolic Weyl groups and the four normed division algebras*, arXiv:0805.3018. <https://arxiv.org/abs/0805.3018>
- [KP] R. Kellerhals, G. Perren, *On the growth of cocompact hyperbolic Coxeter groups*, arXiv:0910.4103. <https://arxiv.org/abs/0910.4103>